

# The Stern Compact-Layer Robin Boundary Condition

Derivation and exact code mapping from Bonnefont–Argoul–Bazant (2001)

PNPInverse / Forward solver

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## Abstract

The PNP–BV forward solver imposes a single Robin boundary condition on the Poisson equation at the electrode,  $\varepsilon_b \partial_n \Phi = C_S (V_{\text{app}} - \Phi)$ , implemented as the residual line

```
F_res -= stern_coeff * (phi_applied - phi) * w * ds(electrode_marker)
```

in `Forward/bv_solver/forms_logc.py:631` and `forms_logc_muh.py:668`. This note derives that line, term by term, from Section 2.2 of Bonnefont, Argoul, and Bazant (2001) [1], the original PNP-with-Faradaic-kinetics treatment in which this exact boundary condition appears coupled to a Butler–Volmer flux at the outer Helmholtz plane.

## 1 The “missing” boundary condition

Once the diffuse layer is resolved as a continuum (rather than collapsed into an effective  $\zeta$ -potential), Poisson’s equation acquires a degree of freedom at the electrode that the species flux conditions do not constrain. Bonnefont, Argoul, and Bazant make this explicit in their Section 2.2 [1, p. 54]:

*“No other boundary conditions are typically mentioned in electrochemistry textbooks because none are needed when the common assumption of electroneutrality is made. In this work, however, since we treat diffuse charge explicitly we also need another boundary condition on the potential. Ignorance of this boundary condition is typically hidden in the ‘zeta-potential’  $\zeta$  ... but in electrochemistry, the zeta potential of an electrode should be determined self-consistently from the microscopic electrochemical boundary conditions.”*

The Stern compact layer supplies this missing condition.

## 2 The Stern boundary condition

Bonnefont *et al.* adopt the Stern model: a thin charge-free dielectric slab of permittivity  $\varepsilon_S$  and effective thickness  $L_S$  between the metal and the outer Helmholtz plane, modelled as a constant areal capacitance

$$C_S = \varepsilon_S / L_S \quad [\text{F/m}^2]. \quad (1)$$

With  $\Delta\Phi_S \equiv \Phi_e - \Phi_r$  denoting the potential drop across the slab (metal minus reaction plane), the paper writes the boundary condition as [1, Eq. (8)]:

$$\Delta\Phi_S = \begin{cases} -\varepsilon_S \partial_X \Phi / C_S & \text{at } X = 0, \\ +\varepsilon_S \partial_X \Phi / C_S & \text{at } X = L, \end{cases} \quad (2)$$

together with the definition of the effective Stern length

$$\lambda_S = \varepsilon_S / C_S, \quad (3)$$

which is [1, Eq. (9)].

### 3 From Eq. (2) to the standard Robin form

Let  $\mathbf{n}$  be the outward normal to the PNP domain  $\Omega$  at the electrode boundary  $\partial\Omega_{\text{el}}$ . At the right electrode  $X = L$ ,  $\mathbf{n} = +\hat{x}$  and  $\partial_n \Phi = \partial_X \Phi$ ; at the left electrode  $X = 0$ ,  $\mathbf{n} = -\hat{x}$  and  $\partial_n \Phi = -\partial_X \Phi$ . Both cases of Eq. (2) then collapse to the single expression

$$\Delta \Phi_S = \lambda_S \partial_n \Phi. \quad (4)$$

Now  $\Delta \Phi_S = V_{\text{app}} - \Phi_{\text{OHP}}$ , where  $\Phi_{\text{OHP}}$  is the boundary value of  $\Phi$  on the diffuse-layer side of the outer Helmholtz plane (the value the PNP solver actually controls). Multiplying both sides of (4) by  $\varepsilon_b / \lambda_S$  gives

$$\boxed{\varepsilon_b \partial_n \Phi = C_{S,\text{eff}} (V_{\text{app}} - \Phi_{\text{OHP}}), \quad C_{S,\text{eff}} \equiv \varepsilon_b / \lambda_S,} \quad (5)$$

which is the Robin boundary condition the solver imposes. Two remarks on the identification of  $C_{S,\text{eff}}$  with the physical compact-layer capacitance:

- In the special case  $\varepsilon_S = \varepsilon_b$ ,  $\lambda_S = L_S$  exactly and  $C_{S,\text{eff}} = \varepsilon_b / L_S = C_S$  literally.
- In the general case  $\varepsilon_S \neq \varepsilon_b$ , the displacement field is continuous across the Stern slab,  $\varepsilon_b \partial_n \Phi|_{\text{diffuse}} = \varepsilon_S \partial_n \Phi|_{\text{Stern}}$ , so  $\varepsilon_b \partial_n \Phi$  on the diffuse side equals  $\sigma_M = C_S \Delta \Phi_S$  (the metal surface charge). The Robin coefficient appearing in (5) is then identified with the literature value of the compact-layer capacitance directly, and the role of  $\lambda_S$  in Eq. (2) is to encode the dielectric mismatch.

For the production target  $C_S = 0.20 \text{ F/m}^2$  ( $L_S \approx 5 \text{ \AA}$ ,  $\varepsilon_S \approx 11.3 \varepsilon_0$ ), see `docs/phase6/CMK3.capacitance.literature`

### 4 Weak form: term-by-term derivation of the residual line

The solver does not implement (5) directly; it substitutes it into the natural-BC slot of the Poisson weak form. We trace each term of the resulting residual to a specific equation in [1].

Multiply the Poisson equation  $-\nabla \cdot (\varepsilon_b \nabla \Phi) = \rho$  (Bonnefont–Argoul–Bazant Eq. (3)) by a test function  $w$  and integrate over  $\Omega$ :

$$-\int_{\Omega} \nabla \cdot (\varepsilon_b \nabla \Phi) w \, dx = \int_{\Omega} \rho w \, dx. \quad (6)$$

Integration by parts (Green's identity) gives

$$\int_{\Omega} \varepsilon_b \nabla \Phi \cdot \nabla w \, dx - \int_{\partial\Omega} \varepsilon_b (\partial_n \Phi) w \, ds = \int_{\Omega} \rho w \, dx. \quad (7)$$

The test space vanishes on the Dirichlet (bulk) boundary, so only the electrode boundary  $\partial\Omega_{\text{el}}$  contributes. On that boundary we substitute (5):

$$\int_{\Omega} \varepsilon_b \nabla \Phi \cdot \nabla w \, dx - \int_{\partial\Omega_{\text{el}}} C_S (V_{\text{app}} - \Phi) w \, ds = \int_{\Omega} \rho w \, dx. \quad (8)$$

Moving the source to the left and identifying the assembled residual  $F_{\text{res}}(\Phi, w)$ ,

$$F_{\text{res}}(\Phi, w) = \underbrace{\int_{\Omega} \varepsilon_b \nabla \Phi \cdot \nabla w \, dx}_{\text{POISSON-VOL}} - \underbrace{\int_{\Omega} \rho w \, dx}_{\text{POISSON-RHS}} - \underbrace{\int_{\partial\Omega_{\text{el}}} C_S (V_{\text{app}} - \Phi) w \, ds}_{\text{STERN}} = 0. \quad (9)$$

## Term-by-term mapping

Each piece of (9) corresponds to one residual assignment in the solver and to one equation in [1]:

| Term        | Math                                                                     | Code (forms_logc.py)                                                                                     | Source in [1]                                                      |
|-------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| POISSON-VOL | $\int_{\Omega} \varepsilon_b \nabla \Phi \cdot \nabla w \, dx$           | <code>F_res += eps_coeff *<br/>dot(grad(phi), grad(w)) *<br/>dx</code> (line 600)                        | Eq. (3), volume<br>term after IBP                                  |
| POISSON-RHS | $-\int_{\Omega} \rho w \, dx$                                            | <code>F_res -= charge_rhs *<br/>sum(z[i]*ci[i]*w) * dx</code><br>(line 602)                              | Eq. (3), RHS<br>source<br>$\rho = zF(C_C - C_A)$                   |
| STERN       | $-\int_{\partial\Omega_{\text{el}}} C_S (V_{\text{app}} - \Phi) w \, ds$ | <code>F_res -= stern_coeff *<br/>(phi_applied - phi) * w *<br/>ds(electrode_marker)</code><br>(line 631) | Eq. (8)<br>substituted into<br>the boundary<br>integral of Eq. (7) |

The Butler–Volmer Faradaic flux ([1, Eq. (7)], applied at the same  $\partial\Omega_{\text{el}}$  with  $\eta = \Delta\Phi_S - E_{\text{eq}} = V_{\text{app}} - \Phi - E_{\text{eq}}$ ) enters the species residuals in the same file at `forms_logc.py:540--590`; this is the second half of the Bonnefont *et al.* boundary system and uses the *same*  $\Phi$  that the Stern term controls.

## Nondimensional form

In dimensionless variables  $\phi = zF\Phi/RT$ ,  $x = X/L$ , Bonnefont *et al.* recast the BC as Eq. (20),  $\Delta\phi_S = \mp \delta_D \varepsilon \partial_x \phi$ , with  $\varepsilon = \lambda_D/L$  and  $\delta_D = \lambda_S/\lambda_D$ . The solver carries the same rescaling internally: see the docstring in `Forward/bv_solver/nondim.py:62--78`, which states verbatim

```
Physical BC: epsilon * grad(phi).n = C_stern * (phi_m - phi)
stern_model = C_stern * V.T / (F * c_ref * L) [dimensionless]
```

i.e. the dimensionless residual is identical to (9) with  $\varepsilon_b$ ,  $\rho$ ,  $C_S$ ,  $V_{\text{app}}$  and  $\Phi$  all replaced by their nondimensional counterparts.

## 5 Limiting cases

Eq. (5) interpolates between two familiar limits:

- $C_S \rightarrow \infty$  (vanishing compact layer): the Robin condition forces  $\Phi \rightarrow V_{\text{app}}$  on  $\partial\Omega_{\text{el}}$ , which is the legacy no-Stern Dirichlet boundary the solver falls back to when `stern_capacitance_fm2` is `None`. In Bonnefont *et al.* notation this is the Helmholtz limit  $\delta_D \rightarrow 0$ .

- $C_S \rightarrow 0$  (perfectly blocking compact layer): the condition forces  $\varepsilon_b \partial_n \Phi \rightarrow 0$ , i.e. no net displacement-current flux into the metal. In [1] this is the “Gouy–Chapman model with  $\delta_D \rightarrow \infty$ .”

Finite  $C_S$  partitions the applied voltage between the compact layer ( $\Delta\Phi_S = \sigma_M/C_S$ ) and the diffuse layer ( $\Phi_{\text{OHP}}$ ), exactly as motivated in [1, §2.2].

## References

- [1] A. Bonnefont, F. Argoul, and M. Z. Bazant. Analysis of diffuse-layer effects on time-dependent interfacial kinetics. *J. Electroanal. Chem.*, 500:52–61, 2001. doi:[10.1016/S0022-0728\(00\)00470-8](https://doi.org/10.1016/S0022-0728(00)00470-8).